AI applications and agents on Azure

1) AI in Azure

1. Which statement best explains the relationship between AI and ML?

AI and ML are interchangeable terms; both refer to systems that mimic human intelligence without distinction.

ML focuses exclusively on generative tasks like creating text and images, whereas AI is limited to decision-making and planning.

AI is the overarching goal of creating systems that exhibit human-like intelligence, while ML is a data-driven method used to achieve AI by learning patterns from data. ✓

2. How does Microsoft Foundry relate to Azure?

Foundry is built on top of Azure and uses Azure resources such as compute, networking, identity, and security to host and operate AI applications. ✓

Foundry runs independently from Azure and doesn't require Azure resources to deploy models or agents.

Foundry replaces Azure services entirely, serving as a standalone cloud platform for running AI workloads.

3. Which statement best describes how keys, secrets, and endpoints work together in an Azure-based AI application?

The key is the location where model responses are stored, and the endpoint retrieves those responses from Azure Key Vault.

The endpoint stores sensitive values for an AI application, and the key determines how much data the endpoint can return.

The endpoint is a URL for calling a deployed model, and the key (stored as a secret in Azure Key Vault) authenticates the request made to that endpoint. ✓

4. Which statement best describes a client application in the context of an AI solution built with Foundry?

A client application is the environment that hosts the model, runs inference, and returns results.

A client application is a program the user interacts with—such as a web app or mobile app—that sends requests to a model endpoint and displays the response. ✓

A client application is a standalone Azure service that automatically generates API keys for model deployments.

2) Text Analysis

1. You need to analyze text where the same input must return structured results based on statistical techniques. Which approach is most appropriate?

The OpenAI responses API, because it can follow natural language instructions.

The Azure Language SDK, because it returns deterministic, structured output. ✓

The chat playground in Foundry, because it supports follow-up questions.

2. What is the purpose of the client object in the Azure Language SDK?

It stores the application's user interface settings.

It helps application code communicate with the Azure Language service. ✓

The client object stores the text that needs to be analyzed.

3. What is the main purpose of the Azure Language MCP server?

To automatically generate website layouts for an agent.

To replace all generative AI models inside an agent.

To expose Azure Language capabilities to agents through the Model Context Protocol. ✓

3) Speech in Azure

1. Why would a developer use the Azure Speech-to-Text SDK instead of only using the Foundry playground?

The SDK replaces the need for Azure Speech models.

The SDK is required to upload audio files to the Foundry portal.

The SDK allows speech recognition to be added directly into application code. ✓

2. What does the Azure Text-to-Speech SDK handle for developers?

Only selecting the voice and writing audio files manually

Authentication, network communication, and audio generation ✓

Storing synthesized audio permanently in Azure Storage

3. What role does the Voice Live Python SDK (azure-ai-voicelive) play in a voice-enabled agent?

It stores audio recordings permanently in Azure Storage

It replaces the need for microphones and speakers on the user's device

It opens a real-time connection, streams audio, and handles spoken responses and interruptions ✓

4) Computer Vision in Azure

1. What is a multimodal model?

A model that can only process images but not text.

A model that can understand and work with more than one type of data, such as text and images. ✓

A model that generates video content only.

2. How can developers programmatically generate images using Foundry image generation models?

By sending text prompts through the OpenAI Responses API using a deployed image model ✓

By uploading images through the Foundry Playground UI.

By calling the GPT-4.1 model endpoint.

3. When you generate images programmatically using the OpenAI Python SDK with Microsoft Foundry, which value should you pass as the model parameter in the request?

The original base model name (for example, gpt-image-1.5).

The deployment name you gave the image generation model in your Foundry resource. ✓

The name you gave your Foundry resource.

4. Why is video generation with Sora models in Microsoft Foundry handled as an asynchronous job?

Because video generation requires user interaction during rendering.

Because the REST API doesn't support synchronous requests.

Because video generation is resource-intensive and takes time to complete. ✓

5) Information Extraction in Azure

1. What is the key advantage of using Azure Content Understanding over basic Optical Character Recognition (OCR)?

Azure Content Understanding extracts text faster by skipping image preprocessing.

Azure Content Understanding understands document structure and maps extracted data to a defined schema. ✓

Azure Content Understanding extracts structured data, while OCR extracts the relationship between words in text.

2. What is the primary role of an analyzer in Azure Content Understanding?

It defines how content is processed and what structured data is returned. ✓

It stores extracted data in a database.

It converts JSON output into human-readable text.

3. When you use the Azure Content Understanding Python SDK, what happens after you submit content for analysis?

The results are returned immediately in the same request.

The analyzer retrains itself on the submitted content.

You must poll a URL until the analysis job completes. ✓

6) Foundry IQ

1. What is the primary purpose of Microsoft Foundry IQ?

To provide a managed knowledge layer that agents can use to retrieve permission-aware enterprise information. ✓

To train a new foundation model from every document in an organization.

To replace AI agents with a traditional keyword search page.

2. What does a Foundry IQ knowledge base contain?

One or more knowledge sources and settings that control retrieval behavior. ✓

Only the conversation history for a single agent.

Only documents uploaded directly to a language model.

3. What does agentic retrieval do with a complex user question?

It can decompose the question, search relevant sources in parallel, rerank results, and return cited evidence. ✓

It permanently adds the question and answer to the model's training data.

It searches every source with an exact keyword match and returns results without ranking.

4. Why should an agent be instructed to cite knowledge base sources?

Citations help users trace and verify the evidence supporting a response. ✓

Citations allow the agent to bypass permissions on the source documents.

Citations cause the language model to retrain itself after every response.

5. Which practice helps protect enterprise information in a Foundry IQ solution?

Use identities and access controls so retrieval returns only content the caller is authorized to access. ✓

Copy every source into one public index so all agents retrieve the same content.

Trust retrieved document text as agent instructions.